# Enterprise Architecture & Risk Assessment Exercise

## Objective

Design a scalable, secure, and highly available enterprise application on Google Cloud, considering the provincial infrastructure in the Philippines and specific architectural choices.

## Background Scenario

You are the Lead Cloud Architect for **"AgriConnect"**, a platform that allows farmers to access market prices, apply for micro-loans, and sell produce directly to distributors.

**User Profile & Constraints:**

- **Tech Stack:** React Native (Mobile), Next.js (Web Frontend), and Nest.js (Backend API).
- **Geographic Distribution:** Users are highly concentrated across provincial Philippines (Luzon, Visayas, Mindanao).
- **Infrastructure Challenge:** Provincial internet connectivity can be intermittent and have high latency to major data centers.
- **Compliance/Security:** PII and financial data must be encrypted and protected under strict data privacy laws.

### Task 1: Software Architecture Diagram

Provide a high-level architecture diagram. Clearly label:

- **Global Edge & Load Balancing:** How users access the system.
- **Compute Tier:** Deployment strategy for Next.js and Nest.js.
- **Data Tier:** The storage and database services.
- **Integration Points:** CDN, asset delivery, and real-time components (chat/live maps).

### Task 2: Architectural Justification

For each component in your diagram, provide a 2–3 sentence explanation focusing on:

1. **Multizone High Availability:** How the architecture tolerates a zone or regional outage.
2. **Provincial Latency Optimization:** How you minimize loading times given the local network constraints.
3. **Budget Optimization:** How the architecture minimizes operating costs without sacrificing availability.

**Task 3: Security & Compliance**

1. **DDoS and OWASP Mitigation:** How you protect external endpoints.
2. **Data Protection:** How you secure data in transit and at rest, implementing the principle of least privilege.

**Task 4: Monitoring, Logging, and Alerting**

1. Explain how you will observe system health and security breaches using GCP services.
2. Define **two specific thresholds/metrics** that will trigger automatic alerts and notifications to the engineering team.

**Task 5: Risk Assessment & Mitigation Matrix**

Fill out the matrix below:

| **Risk Category** | **Specific Risk Description** | **Likelihood (High/Med/Low)** | **Impact (High/Med/Low)** | **Mitigation Strategy on GCP** |
|---|---|---|---|---|
| **Connectivity** | High latency for provincial users during peak hours. | | | |
| **Availability** | Zonal infrastructure failure affecting operations. | | | |
| **Security** | Data breach of PII in transit or at rest. | | | |
| **Budget** | Unexpected scaling costs from sudden traffic spikes. | | | |